SOC Lab Investigation #001

Title: Failed Logon Attempts Followed by a Successful Logon

Date: October 9, 2026
Environment: Windows Virtual Machine (VirtualBox)
Log Source: Windows Security Event Log

Objective

Investigate Windows authentication events and determine whether multiple failed logon attempts were followed by a successful logon.

Evidence Collected

- **Event ID 4625:** Failed logon attempts recorded at 12:08:01 and 12:08:03.
- **Status:** 0xC000006D
- **Sub Status:** 0xC000006A — Incorrect password.
- **Logon Type:** 2 — Interactive logon.
- **Event ID 4624:** Successful logon recorded at 12:08:05.
- **Account:** Local Windows test account.
- **Successful Logon Type:** 2 — Interactive logon.

Analysis

The Windows Security logs showed two failed authentication attempts followed by a successful interactive logon for the same local account.

The failed attempts were caused by incorrect passwords. The activity was intentionally generated in an authorized laboratory environment.

Conclusion

Classification: Authorized Lab Activity

Security Incident: No confirmed compromise.

The observed authentication sequence demonstrates why failed logon events must be analyzed in context rather than automatically classified as brute-force attacks.

Evidence & Screenshots

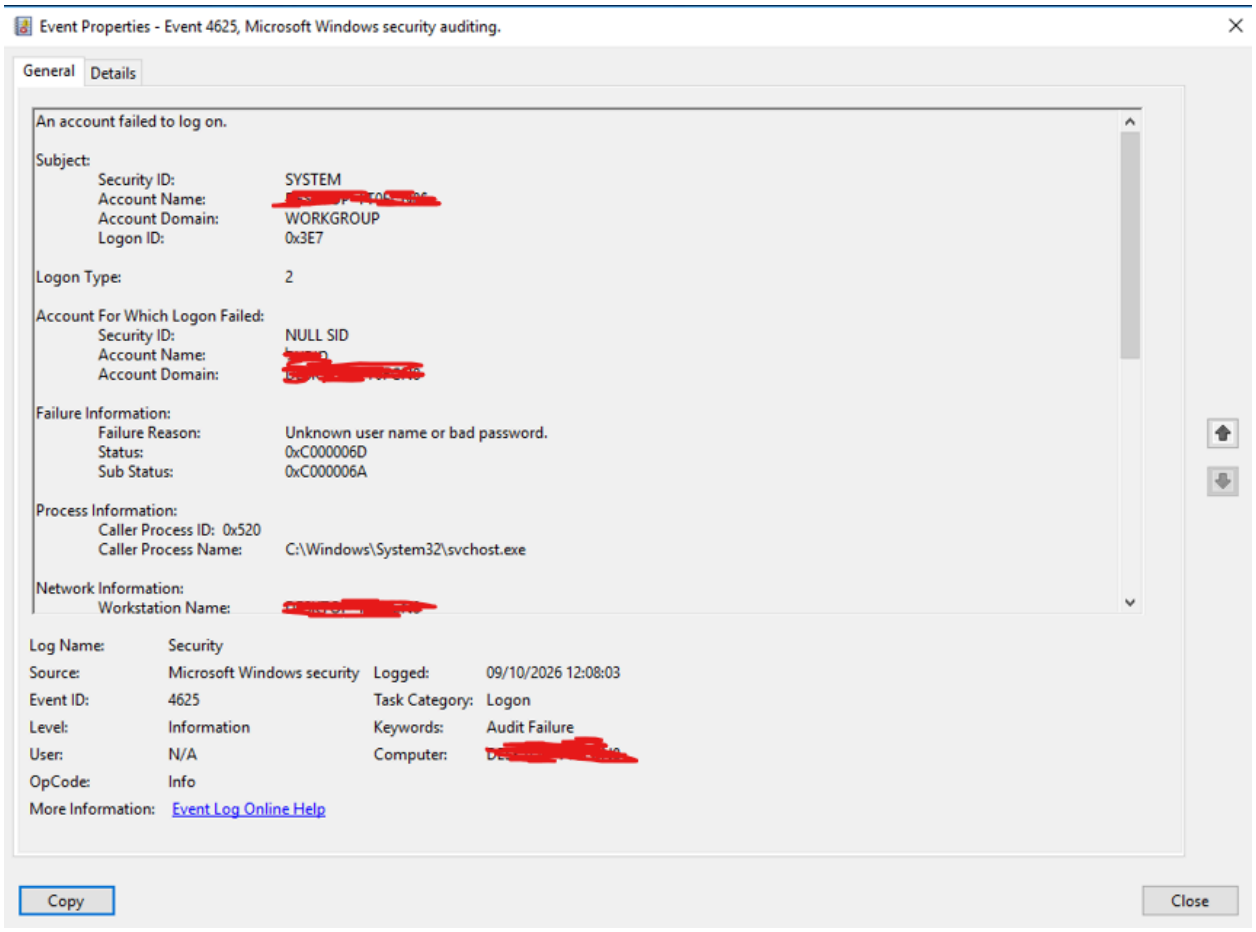


Figure 1 — Failed Logon (Event ID 4625)

Windows Security Event Log showing a failed interactive logon attempt. The event contains Sub Status 0xC000006A, indicating an incorrect password for an existing account.

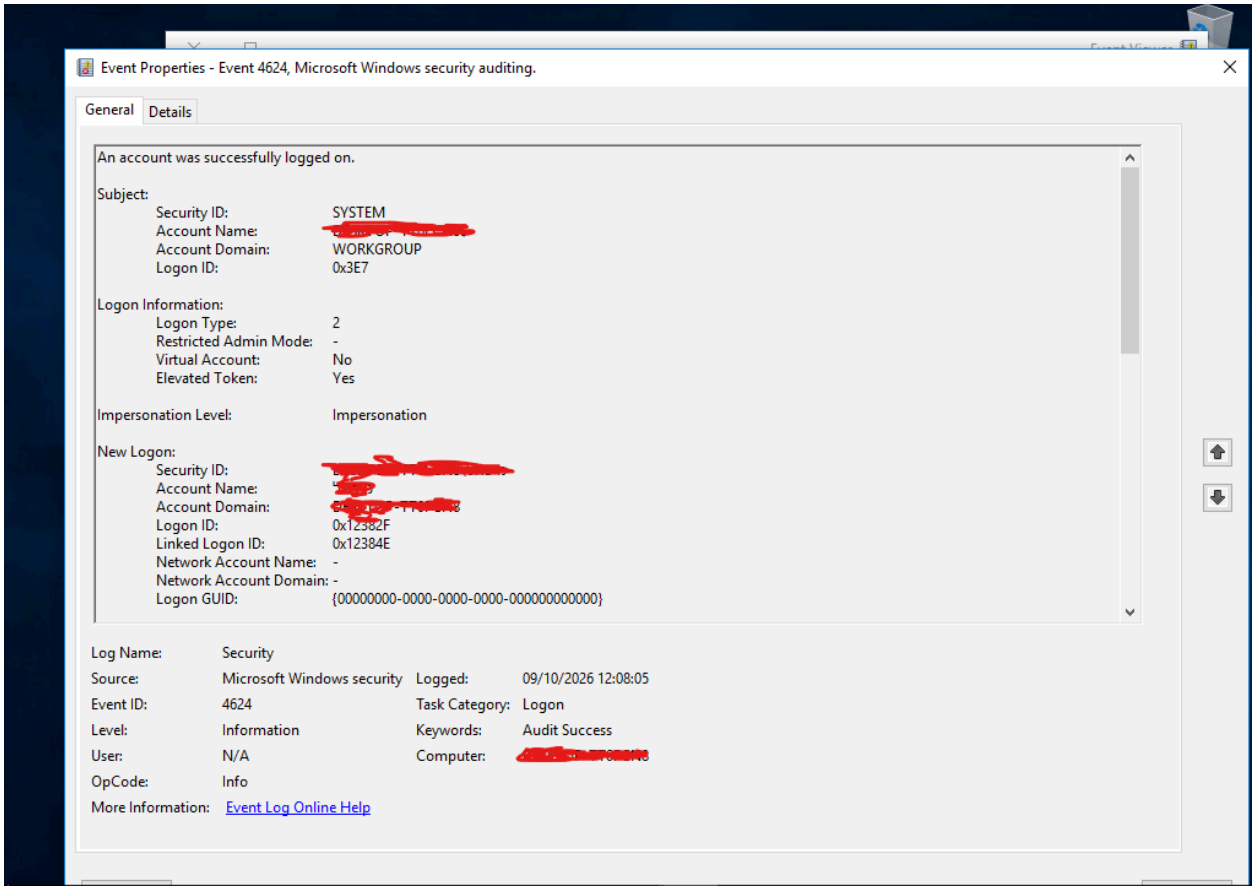


Figure 2 — Successful Logon (Event ID 4624)

Windows Security Event Log showing a successful interactive logon at 12:08:05, shortly after the failed authentication attempts.